

Machine Learning Safety

Exercise Sheet - 06

Exercise 6.1: (Optional)

* Advantages of explainable models in safety critical systems.

- ① Improved trust and transparency because, we can understand why a model made a decision.
- ② Safety certification ensures that the model ~~is~~ ⁱⁿ behaves safely and fairly, especially in domains like autonomous driving and healthcare.
- ③ Explainability can reveal bias and error by detecting when the model relies on incorrect features rather than actual objects.
- ④ We can identify weakness in training data and improve robustness.
- ⑤ We can verify whether system's reasoning appears reasonable before acting on predictions.

* Disadvantages of current explainability methods.

- ① Explanations may be unfaithful:
convincing explanation may not reflect model's true reasoning.
- ② Correlation vs Causation:
Methods may highlight correlated features rather than causal features.
- ③ Diff methods like Saliency, GradCam, Occlusion can produce diff ~~output~~ exp. for same prediction.

- ④ Risk of Overtrust
 ⑤ Limited evidence: Explanations provide evidence about model behavior but does not guarantee that model is correct.

Exercise 6.2 Local vs Global Exp

- ① Local Exp: (understand decision process for a single input)
- * Aims to explain why the model made a prediction for a specific input
 - * A method like Grad CAM (Class Activation Maps) highlights image regions that were most relevant for classification.
 - * It answers the question, "Which part of this image were responsible for this prediction?"
- ② Global Exp: (understand decision process in general)
- * Aims to under how model makes
 - * A method used in GExp is "Filter Visualization" which finds input patterns that maximally activate a "convolutional filter."
 - * It answers question like "What kind of features has the model learned and what do its filters detect?"

* CAM → take AM for input x^c
 * Weight each AM A_k by its weight w_k^c
 * Sum weighted maps → $\sum_k w_k^c A_k$
 * Scale up to input size.

* Grad CAM → $\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{dy^c}{dA_{ij}^k}$ | $L^c_{GC} = \text{ReLU}(\sum_k \alpha_k^c A_k^c)$

Exercise 6.3:

① Occlusion: Works by covering (occluding) parts of input image and measuring how class probability changes. If occluding a region decreases pred then that region is considered important

* Adv over Saliency:

- ① Easy to understand
- ② Can be applied to various models.

* Disadv: (intuitive but slow)

- ① Computationally expensive, especially for high-resolution maps. Because, many occluded versions of image must be evaluated.
- ② Output maps tends to have low resolution.

② Saliency Maps: Work by computing gradient of the class score wrt input pixels.

Gradient indicates how sensitive the pred is to changes in each pixel.

* Adv over Occlusion:

- ① Efficient to compute using gradients and auto differentiation.

* Disadv: (fast but noisy)

- ① Produces noisy pixel-level explanation that can be difficult to ~~exp~~ interpret visually

Exercise 6.4: Chain of Thought Fidelity

① A thinking trace is faithful if it accurately reflects a model's actual internal reasoning process that led to the answer.

* Faithfulness is hard to verify because, we only observe generated exp, not the model's true internal computation.

* A model may produce a really good explanation that differs from its actual reasoning.

② Simulatability: given thinking trace, can an outside observer predict the model's answer?

Eg: Question: Will this loan application be approved?

Ans: Approved.

Thinking trace: Applicant has a good credit score, low debt and steady job, so the loan should be approved.

Why simulatability is satisfied: a person reading the trace can easily predict the output as "approved".

Reality/ why trace is unfaithful:

Model did not really check those factors.

It learned from correlations in training data like income range, age, repayment history.

The explanation ~~is~~ does not reflect the true causal features used by the model.